Numerical Integration and the Standard Normal CDF

Exercises 1–2: Riemann/Lebesgue viewpoints and numerical methods

Abstract

This report compares two numerical interpretations of integration. A Riemann-type method partitions the domain and samples the function on each subinterval. A Lebesgue-style method partitions the range of function values and estimates the measures of the corresponding preimages. For bounded Riemann-integrable functions, both approximations converge to the same integral when the domain grid and range grid are refined. Examples with a jump, an integrable singularity, Thomae's function, and the Dirichlet function show where convergence is slow or a finite sampler is fundamentally insufficient. The second exercise implements the standard normal CDF using Simpson quadrature, a power series, and an error-function approximation, and compares the results with published reference values.

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1 Exercise 1 – Riemann and Lebesgue integration

1.1 Two ways to approximate an integral

Let $f : [a, b] \to \mathbb{R}$ be bounded. Divide the domain into N equal intervals of width

$$\Delta x = \frac{b-a}{N}, \qquad x_i = a + \left(i + \frac{1}{2}\right) \Delta x.$$

The midpoint Riemann sum is

$$R_N(f) = \Delta x \sum_{i=0}^{N-1} f(x_i). \tag{1}$$

This is a Riemann sum: the domain is partitioned, and one representative value is selected in each subinterval.

For the Lebesgue-style construction, partition the range into B bins of width $\Delta y = (M - m)/B$, where $m \le f(x) \le M$:

$$I_j = [m + j\Delta y, m + (j + 1)\Delta y), \qquad A_j = f^{-1}(I_j) = \{x : f(x) \in I_j\}.$$

A simple-function approximation uses a representative value in each range bin. With the midpoint representative, its integral is

$$L_{N,B}(f) = \sum_{j=0}^{B-1} \left(m + \left(j + \frac{1}{2}\right) \Delta y\right) \widehat{\mu}(A_j), \tag{2}$$

where the preimage measure is estimated from the same domain grid:

$$\widehat{\mu}(A_j) = \Delta x \, \#\{i : f(x_i) \in I_j\}.$$

Thus the program does not calculate arbitrary preimage measures exactly. It builds a histogram of sampled function values, which is a numerical approximation to the distribution of values of f over the domain.

For an exact measure calculation, the lower and upper simple functions give

$$\sum_j (m + j\Delta y) \operatorname{Leb}(A_j) \le \int_a^b f(x) \, \mathrm{d}x \le \sum_j (m + (j + 1)\Delta y) \operatorname{Leb}(A_j),$$

with a gap of at most $(b - a)\Delta y$. The program reports the corresponding sample-based lower and upper histogram sums as well as their midpoint. Since the measures are themselves estimated from a finite grid, this band is a range-quantization band, not a rigorous bound for the true integral.

1.2 Functions and analytical values

The experiment uses the following examples.

Function	Relevant property	Integral / status
$\sin x$, $x \in [0, \pi]$	Continuous and bounded	2
x^2, $x \in [0, 1]$	Continuous and bounded	1/3
1 for $x < 1/2$, 3 otherwise	One jump discontinuity	2
$\lvert x - 1/2 \rvert$, $x \in [0, 1]$	Continuous, one corner	1/4
$1/\sqrt{x}$, $x \in (0, 1]$	Unbounded at 0; improper integral converges	2
1 for $x < 1/2$, 0 otherwise	Indicator of an interval	1/2
$\sin x/x$, $x \in [0, \pi]$	Define $f(0) = 1$ by continuity	$\mathrm{Si}(\pi) \approx$ 1.851937052
Thomae's function	Dense discontinuities, but Riemann-integrable	0
Dirichlet's function	Discontinuous everywhere; not Riemann-integrable	Lebesgue integral 0

For the sinc example,

$$\int_0^{\pi} \frac{\sin x}{x}\,\mathrm{d}x = \mathrm{Si}(\pi), \qquad \mathrm{Si}(u) = \int_0^{u} \frac{\sin t}{t}\,\mathrm{d}t.$$

The value is quoted numerically because the sine integral is itself a special function.

1.3 Numerical results

The practical implementation uses $N = 10{,}000$ midpoint samples and $B = 2{,}000$ range bins. For $1/\sqrt{x}$, the range is truncated at $Y_{\max} = 32$. The table reports the domain midpoint sum and the midpoint of the sample-based range histogram.

Function	Exact / theoretical value	Riemann midpoint	Range-bin midpoint
$\sin x$	2	2.0000000082	1.9999953744
x^2	0.3333333333	0.3333333325	0.3333355000
Step $(1, 3)$	2	2.0000000000	2.0000000000
$\lvert x - 1/2 \rvert$	0.25	0.2500000000	0.2500113500
$1/\sqrt{x}$	2	1.9939510138	1.9687152000
Indicator $(1, 0)$	0.5	0.5000000000	0.5000000000
$\sin x/x$	1.8519370520	1.8519370533	1.8519332122
Thomae	0	0.0002100000	0.0003700000
Dirichlet	0 (Lebesgue); no Riemann integral	1.0000000000	0.9997500000

The last row is deliberately not interpreted as convergence to the Lebesgue integral. The values sampled by JavaScript are finite binary floating-point numbers, and every such number is rational. Consequently, the grid evaluator of the Dirichlet function returns 1 at every sample. The exact measure-theoretic calculation is different:

$$D^{-1}(\{1\}) = \mathbb{Q} \cap [0, 1], \qquad D^{-1}(\{0\}) = [0, 1] \setminus \mathbb{Q},$$

so that

$$\int_0^1 D(x)\,\mathrm{d}x = 1 \cdot \mathrm{Leb}(\mathbb{Q} \cap [0, 1]) + 0 \cdot \mathrm{Leb}([0, 1] \setminus \mathbb{Q}) = 0.$$

The rationals are countable and therefore have Lebesgue measure zero. On every nonempty interval, however, both rationals and irrationals are dense. The lower Darboux sum is always 0 and the upper Darboux sum is always 1, so the function is not Riemann-integrable. A finite histogram cannot recover the exact preimage measure unless that measure-theoretic information is supplied separately.

For Thomae's function,

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \\ 0, & x \notin \mathbb{Q}, \end{cases}$$

the integral is 0. For each $\varepsilon > 0$, the set where $T(x) \geq \varepsilon$ contains only fractions whose reduced denominator satisfies $q \leq 1/\varepsilon$; there are only finitely many such fractions in $[0, 1]$. Thus the set of points where T is at least ε has measure zero, and the function is Riemann-integrable with integral zero. Its discontinuities are dense, but they form a measure-zero set. Midpoint grids use rational points and produce positive values, although the sums tend to zero as the grid is refined.

1.4 Interpretation: convergence and computational cost

For bounded Riemann-integrable functions, the Riemann sums converge to the common Riemann/Lebesgue integral as the domain mesh tends to zero. The range-bin method also converges when both discretizations are refined: $N \to \infty$ improves the estimates of the preimage measures, while $B \to \infty$ reduces the range quantization width. A fixed B leaves a quantization error even if N is extremely large.

The examples show different convergence behavior:

- For smooth functions such as $\sin x$ and x^2, midpoint sums converge quickly (second order in the mesh for sufficiently smooth functions). The range-bin estimate has an additional error of order Δy.
- For a step function, the domain discretization error is normally of order $1/N$ near the jump. In the current experiment, the jump is at $1/2$ and the even grid is aligned symmetrically, so the midpoint sum is exact up to floating-point rounding.
- For $1/\sqrt{x}$, the midpoint rule avoids evaluating the singular endpoint but converges more slowly. This function is not bounded Riemann-integrable on the closed interval in the proper sense, but its improper Riemann integral exists and agrees with its Lebesgue integral. For the range method, $f_M = \min(f, M)$ satisfies

  $$\int_0^1 f_M(x)\,\mathrm{d}x = 2 - \frac{1}{M}, \qquad \int_0^1 (f - f_M)\,\mathrm{d}x = \frac{1}{M}.$$

  Therefore the cap M must also tend to infinity. In the table, the omitted tail is $1/32 = 0.03125$ and the range-bin width is $32/2000 = 0.016$.
- Thomae's function is integrable but convergence from rational samples is comparatively slow. The Dirichlet function is not Riemann-integrable, and grid sampling cannot reproduce its true Lebesgue integral.

For a direct implementation, the Riemann midpoint sum takes $O(N)$ time and $O(1)$ auxiliary memory. The histogram method takes $O(N + B)$ time and $O(B)$ memory: it evaluates the same N sample points, assigns them to bins, and then sums the bin contributions. It therefore has a larger constant cost for ordinary numerical functions. Its conceptual advantage appears when the preimage measures are known or can be computed efficiently; it is not automatically more accurate when those measures are estimated by sampling.

2 Exercise 2 – Numerical computation of the standard normal CDF

2.1 Definition and symmetry

The standard normal density and cumulative distribution function are

$$\varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \qquad \Phi(z) = \int_{-\infty}^{z} \varphi(t)\,\mathrm{d}t.$$

The density is even and has total integral one. Therefore

$$\Phi(0) = \frac{1}{2}, \qquad \Phi(-z) = 1 - \Phi(z), \tag{3}$$

and, for finite z,

$$\Phi(z) = \frac{1}{2} + \operatorname{sgn}(z) \int_{0}^{|z|} \varphi(t)\,\mathrm{d}t.$$

This finite-interval form is convenient for numerical integration.

2.2 Why the antiderivative is not elementary: Liouville's theorem

The relevant distinction is that the Gaussian density does have an antiderivative, but no antiderivative expressible by elementary functions. The supplied Hogg–McKean–Craig notes make this distinction explicitly and then derive a power-series antiderivative by term-by-term integration [1].

A standard consequence of Liouville's theorem on elementary integration is the following criterion for this exponential integrand: if $e^{g(x)}$ has an elementary primitive in the differential field generated by x and $e^{g(x)}$, then a rational function $R(x)$ must satisfy

$$R'(x) + g'(x)R(x) = 1.$$

For $g(x) = -x^2/2$, this becomes

$$R'(x) - xR(x) = 1. \tag{4}$$

There is no rational solution. Indeed, a pole of order $m \geq 1$ in R would create a pole of order $m+1$ in R', while xR has at most a pole of order m; the leading pole cannot cancel. Hence R would have no finite poles and would have to be a polynomial. If R is a nonzero polynomial of degree d, then $-xR$ has degree $d+1$, while R' has degree $d-1$, so their difference cannot be the constant 1. The zero polynomial also fails. Thus an elementary primitive is impossible. A detailed differential-algebra treatment of Liouville's theorem is given by Churchill [5].

2.3 The error function gives an exact special-function formula

The error function is defined by

$$\operatorname{erf}(u) = \frac{2}{\sqrt{\pi}} \int_{0}^{u} e^{-t^2}\,\mathrm{d}t, \qquad \operatorname{erfc}(u) = 1 - \operatorname{erf}(u).$$

With the substitution $t = \sqrt{2}\,s$, the normal CDF becomes

$$\boxed{\Phi(z) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{z}{\sqrt{2}}\right)\right] = \frac{1}{2}\operatorname{erfc}\left(-\frac{z}{\sqrt{2}}\right).} \tag{5}$$

Equation (5) is an exact analytical representation, but erf is a special function rather than an elementary one. Numerical computation is still required to return a decimal value. Numerical libraries implement the special function using approximations, series, rational functions, continued fractions, or quadrature. The SciPy documentation defines the same ndtr integral and records, for example, $\operatorname{ndtr}(0.5) = 0.6914624612740131$ [3].

2.4 Numerical method A: composite Simpson integration

For an even integer n, let $h = |z|/n$ and $g_i = \varphi(ih)$. The composite Simpson approximation is

$$I_n(z) = \frac{h}{3}\left[g_0 + g_n + 4\sum_{\substack{1\le i<n\\ i\ \text{odd}}} g_i + 2\sum_{\substack{2\le i<n\\ i\ \text{even}}} g_i\right].$$

Then

$$\Phi_n(z) = \frac{1}{2} + \operatorname{sgn}(z) I_n(z). \tag{6}$$

For a sufficiently smooth integrand, Simpson's global quadrature error is $O(h^4)$; the cost is $O(n)$ density evaluations. The HTML implementation uses $n = 256$ for its comparison table.

2.5 Numerical method B: integrated power series

Starting from the exponential series,

$$e^{-t^2/2} = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{2^k k!},$$

term-by-term integration on a finite interval gives

$$\Phi(z) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2^k (2k+1) k!}. \tag{7}$$

This is the alternating series presented in the supplied notes [1]. To avoid repeatedly computing powers and factorials, write

$$a_0 = |z|, \qquad a_{k+1} = -a_k \frac{z^2(2k+1)}{2(k+1)(2k+3)}.$$

The terms are accumulated until the terms are sufficiently small after the sequence has begun decreasing. Symmetry (3) is used for negative arguments. The series converges for every finite z, but direct summation becomes numerically unattractive for large $|z|$ because of cancellation between large intermediate terms. The implementation therefore uses it only for moderate arguments (here $|z| \le 3.5$).

2.6 Numerical method C: Abramowitz–Stegun error-function approximation

Formula 7.1.26 of Abramowitz and Stegun [4] approximates the error function for $x \ge 0$ by

$$\operatorname{erf}(x) \approx 1 - e^{-x^2} P(t), \qquad t = \frac{1}{1 + px},$$

where

$$\begin{aligned} p &= 0.3275911, \\ P(t) &= a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5, \\ (a_1, a_2, a_3, a_4, a_5) &= (0.254829592, -0.284496736, 1.421413741, -1.453152027, 1.061405429). \end{aligned}$$

The published absolute error bound for this approximation is 1.5×10^{-7} for erf [4]. Since the CDF uses one half of the error-function value, the corresponding absolute error in Φ is bounded by approximately 7.5×10^{-8} (apart from floating-point rounding).

For better tail stability, the code evaluates the complementary error function at a nonnegative argument. If $u = |z|/\sqrt{2}$, then

$$\widehat{q}(u) = \frac{1}{2} e^{-u^2} P\left(\frac{1}{1+pu}\right), \qquad \widehat{\Phi}(z) = \begin{cases} 1 - \widehat{q}(u), & z > 0, \\ 1/2, & z = 0, \\ \widehat{q}(u), & z < 0. \end{cases}$$

The polynomial is evaluated in Horner form. This requires a fixed number of arithmetic operations and one exponential evaluation, so the method has $O(1)$ work per input. For a positive input close to the upper tail, the survival probability $1 - \Phi(z)$ is available directly as $\widehat{q}(u)$ rather than by subtracting two nearly equal numbers.

2.7 Numerical comparison with a published reference

The NIST/SEMATECH e-Handbook publishes the standard-normal table to five decimal places and explains the symmetry relation for negative arguments [2]. The following table compares the three implementations against those tabulated values. Numerical columns are shown to nine decimal places; the reference is limited to five decimal places by the source table.

z	A&S erf	Simpson ($n = 256$)	Power series	NIST reference
−3.00	0.001349967	0.001349898	0.001349898	0.00135
−2.00	0.022750063	0.022750132	0.022750132	0.02275
−1.00	0.158655264	0.158655254	0.158655254	0.15866
−0.50	0.308537537	0.308537539	0.308537539	0.30854
0.00	0.500000000	0.500000000	0.500000000	0.50000
0.50	0.691462463	0.691462461	0.691462461	0.69146
1.00	0.841344736	0.841344746	0.841344746	0.84134
1.50	0.933192769	0.933192799	0.933192799	0.93319
2.00	0.977249937	0.977249868	0.977249868	0.97725
3.00	0.998650033	0.998650102	0.998650102	0.99865

The NIST values agree with the implementations to the precision available in that table. As a higher-precision spot check, SciPy's online documentation gives $\mathrm{ndtr}(0.5) = 0.6914624612740131$ [3]; the A&S implementation gives 0.691462462724 at that point, a difference of about 1.45×10^{-9}. A separate double-precision comparison over the tabulated points gives a maximum absolute difference of about 7×10^{-8} for the A&S approximation, below its stated bound; Simpson with 256 subintervals is within about 9×10^{-12}, while the power series is at approximately double-precision roundoff for these moderate arguments. The finite-precision reference table itself is rounded, so it cannot validate errors smaller than roughly 5×10^{-6} on its own.

2.8 Efficiency and limitations

The three methods make different trade-offs.

- **A&S approximation:** fixed work per argument, low memory use, and a documented absolute error below about 7.5×10^{-8} for the CDF. It is the fastest of the three in the browser benchmark included in the HTML file.
- **Simpson integration:** work proportional to the chosen number of panels. Its error decreases rapidly for this smooth integrand, and the method is easy to generalize to other integrals. A larger panel count costs more function evaluations.

- **Power series:** work proportional to the number of terms required for the tolerance. For moderate arguments, it converges quickly and is very accurate. In the far tails, cancellation makes direct summation less stable; a complementary-error-function or continued-fraction method is preferable.

The HTML page contains an interactive z input, a table of reference values, and a browser-side benchmark. Timings vary by browser, computer, and current system load, so no machine-specific runtime is presented as a universal result. In practice, production libraries use carefully chosen piecewise rational approximations and tail-specific algorithms rather than relying on one Taylor series for all arguments.

3 Conclusions

For ordinary bounded Riemann-integrable functions, refining both the domain partition and the range partition makes the Riemann and Lebesgue-inspired numerical estimates converge to the same integral. The range-based formulation is conceptually different, but when its preimage measures are obtained from the same finite sample grid it still inherits the limitations of sampling. Thomae's function shows that convergence can be slow even when both integrals exist; the Dirichlet function shows that sampling cannot replace exact measure-theoretic information. The unbounded function $1/\sqrt{x}$ adds a truncation error that must be controlled separately.

For the normal CDF, the special function erf gives an exact analytical representation even though the defining Gaussian integral has no elementary antiderivative. Numerical integration, power series, and rational approximations then provide practical decimal values. The error-function approximation is the most economical method in the implementation, while Simpson integration and the integrated series are useful independent methods for checking accuracy and understanding convergence.

References

[1] R. V. Hogg, J. W. McKean, and A. T. Craig, *Introduction to Mathematical Statistics*, Section 3.4, online course notes, East Tennessee State University, revised July 21, 2021, pp. 5–6. https://faculty.etsu.edu/gardnerr/4047/notes-Hogg-McKean-Craig/Hogg-McKean-Craig-3-4.pdf

[2] NIST/SEMATECH, *e-Handbook of Statistical Methods*, "Cumulative Distribution Function of the Standard Normal Distribution," Section 1.3.6.7.1. https://www.itl.nist.gov/div898/handbook/eda/section3/eda3671.htm

[3] The SciPy community, *scipy.special.ndtr – Cumulative distribution of the standard normal distribution*, SciPy Reference Guide. https://docs.scipy.org/doc/scipy/reference/generated/scipy.special.ndtr.html

[4] M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*, National Bureau of Standards, Applied Mathematics Series 55, 1964, p. 299, formula 7.1.26.
Digitized page: https://digital.library.unt.edu/ark:/67531/metadc40302/m1/315/ocr/

[5] R. C. Churchill, *Liouville's Theorem on Integration in Terms of Elementary Functions*, lecture notes for the Kolchin Seminar on Differential Algebra, 2006. https://ksda.ccny.cuny.edu/PostedPapers/liouv06.pdf